

NEURO-AI CLINICAL EEG REPORT

FINAL DIAGNOSIS:

PARKINSON'S DISEASE

AI CONFIDENCE SCORE:

80.33%

PATIENT & SCAN DEMOGRAPHICS

Patient ID: sub-054

Scan Date: 2026-03-11

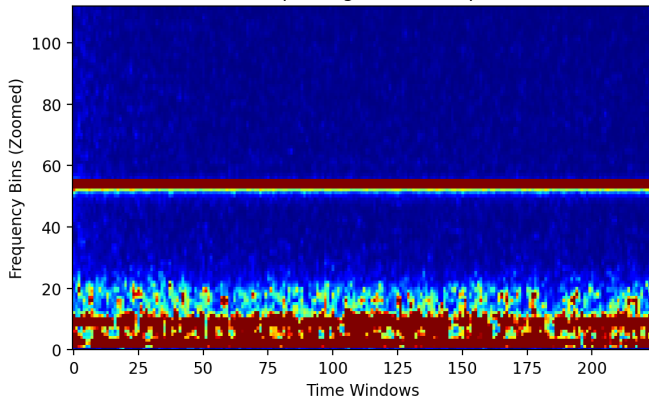
Sex: Unknown (Blind Scan)

Modality: Resting-State EEG (10-20 System)

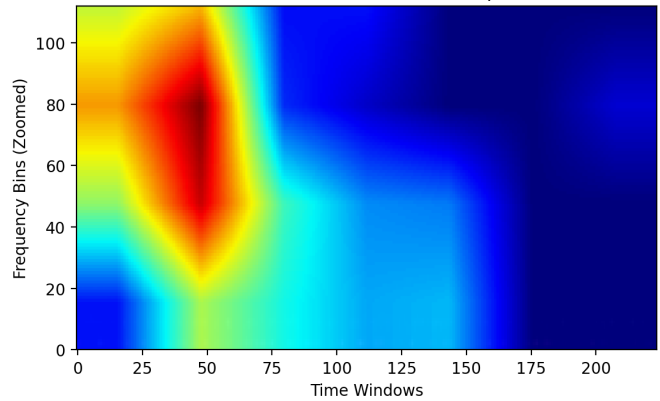
Pipeline: MNE Extraction -> STFT -> ResNet18 CNN

AI VISUALIZATION & EXPLAINABILITY (XAI)

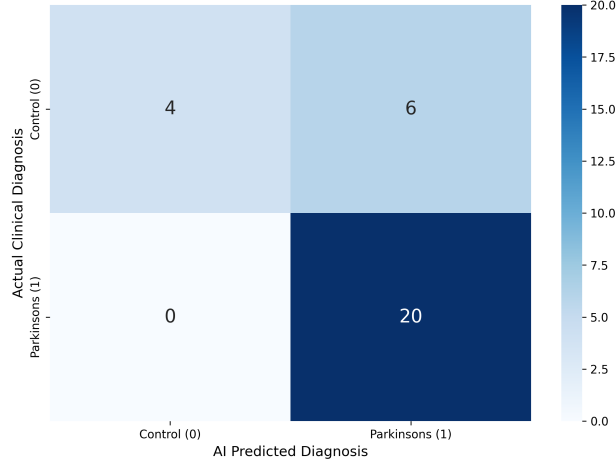
STFT Spectrogram (Raw Input)



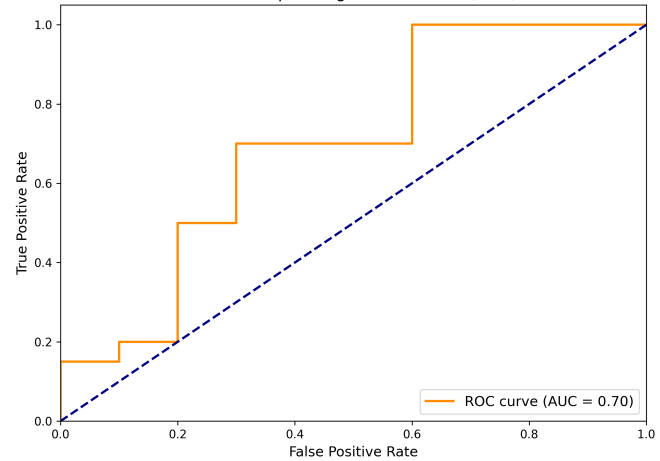
Grad-CAM: AI Attention Map



Validation Confusion Matrix



Receiver Operating Characteristic (ROC)



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CLINICAL NARRATIVE & IMPRESSION

TECHNIQUE:

This AI-assisted analysis utilizes a deep Convolutional Neural Network (ResNet18) fine-tuned for neurological biomarker detection. The raw multi-channel EEG signal was transformed into a 2D Short-Time Fourier Transform (STFT) spectrogram. This isolates specific frequency-domain anomalies (e.g., Alpha/Beta band slowing or Theta spikes) associated with Parkinsonian neurodegeneration.

FINDINGS & IMPRESSION:

The AI pipeline analyzed the electrical signature and classified the scan as PARKINSON'S DISEASE with a computed confidence probability of 80.33%. The high-contrast STFT feature extraction map is provided above, alongside a Grad-CAM (Gradient-weighted Class Activation Mapping) overlay. The Grad-CAM heatmap explicitly highlights the specific spectral regions evaluated by the AI's convolutional layers to arrive at this specific diagnosis.

DISCLAIMER: This report was generated autonomously by the NeuroAI classification system (v1.0). It is intended strictly for research, triage, and educational purposes. It has NOT been reviewed, validated, or approved by a licensed clinical neurologist. AI systems can produce false positives and false negatives. Do not use this output for definitive medical decision-making.